Pattern Recognition Letters 172 (2023) 82–88

Contents lists available at ScienceDirect

Pattern Recognition Letters

journal homepage: www.elsevier.com/locate/patrec

Multi-scale attention and dilation network for small defect detection☆

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ARTICLE INFO

Article history:
Received 22 February 2023
Revised 6 April 2023
Accepted 7 June 2023
Available online 8 June 2023

Edited by: Jiwen Lu

Keywords:
Small object detection
Dilated convolution blocks
Convolutional block attention module (CBAM)
Global context information
Perceptual field

ABSTRACT

Although object detection methods have got surprising performance in simple natural scenes, it is still a challenging task to apply them to complex scenes, especially when the detected objects are small which is common in defect detection tasks of industrial scenarios. Motivated by the higher resolution feature maps, the better detection performance on small objects, we increase the resolution of the feature layers in this paper, and introduce convolutional block attention module (CBAM) into the feature aggregation network to better integrate features at different scales and make better use of the information of small defects. For solving the problem of global context information loss caused by limiting the perceptual field by increasing the resolution of the feature layers, we use dilated convolution blocks to expand the perceptual field. Experimental results show that our method improves the mAP(0.5) on the Industrial Surface Defect Dataset from 88.3% in Yolov5 to 89.2%, mAP(0.75) on the DAGM 2007 increases from 76.56% in Yolov5 to 77.65%, mAP on small object increases by 2.6% on Industrial Surface Defect Dataset and 5.8% on DAGM 2007, which proves the effectiveness of our method for small defect detection.

1. Introduction

In industrial visual tasks such as defect detection [1,2] and defect segmentation [3,4], defects in a singel view are often at multi-scales, which means there are both small and large defects, how to process multi-scale defects has become an important research topic for industrial visual tasks. For the detection and recognition of defects at different scales, it is especially important to be able to perceive information at different scales. Using multi-scale features to represent multi-scale defects has achieved impressive performance on visual tasks, and the multi-scale feature layer has become an essential component of the defect detection algorithm. Specifically, the detection of large-scale defects often requires us to stack convolution operators to obtain a larger receptive field, while small-scale defects need us to process on a high-resolution feature scale with a smaller perceptual field. Industrial vision tasks are a type of visual tasks, so how to build an effective multi-scale feature layer has become a key factor in determining the performance of industrial visual tasks like defects detection.

In recent years, many methods for constructing multi-scale features have proven effective [5–8]. The most common method is FPN [5]. FPN upsamples more abstract and semantic high-level feature maps from top to bottom, and then fuses the upsampling results with feature maps of the same size generated from the bottom-up path through horizontal connection to build multi-scale features with strong representation capabilities. PANet [6] enhances FPN through bidirectional feature fusion. PANet [6] adds a bottom-up path augmentation to FPN [5], which greatly enhances the feature aggregation ability of FPN [5]. There are also ways to introduce attention mechanisms [9] and contextual information into FPNs [7,8], all of which have achieved good results. The MS COCO dataset [10] defines a small object as an object with a resolution less than 32 squared pixels, which means small objects have low resolution and limited pixels, and is hard to get effective feature representation by feature extractor. Although state-of-the-art object detection has shown impressive performance in natural scenes, it is still challenging to detect small defects. In defect detection tasks of industrial scenarios, most of the objects in one picture are small. Therefore, improving the performance of the detector on small defects is an important task in industrial defect detection.

This paper will discuss how to construct multi-scale features with strong representation capabilities for small defects. We

☆ This work was supported by the National Key Research and Development Program of China under Grant 2020AAA0108302 and the Fundamental Research Funds for the Central Universities under Grant xtr072022001.

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https://doi.org/10.1016/j.patrec.2023.06.010
0167-8655/

mainly improve the traditional method from the feature aggregation network. We focus on the construction of feature layers and propose a simple and efficient multi-scale feature layer of small defects specialization, and we call it multi-scale attention and dilation feature layer, to solve the difficulty of small defect detection. Our method can not only perceive the finer and more local features of small defects on the smaller perception field, but also compensate for the global information lost by the small perception field through context aggregation. Experiments have shown that our proposed multi-scale attention and dilation feature layer can significantly improve the detection performance of small defects. To evaluate the performance of the proposed model, we conduct experiments on DAGM 2007 [11] and Industrial Surface Defect Dataset, and compared with baseline.

In summary, the main contributions of this paper are as follows:

1) High-resolution features with attention mechanisms are built which can better perceive the information of small defects.
2) Dilated convolution is used to solve the resolution limitation problem caused by high resolution, which is able to aggregate context of small defects features perceived in a smaller window.
3) Experimental results show that our method gets better performance in small defects detection compared with baseline YoloV5.

2. Related work

Most of the mainstream defect detection algorithms are based on deep learning, including anomaly detection [12–14], image reconstruction [15–19], and object detection [2,7,20–23]. Chun-Liang Li et al. [12] proposed a self-supervised method to learn the normal mode of samples, first cuts a small rectangular region from normal data and pastes it at random location, and then use a one-class classifier to determine whether the image contains anomalies, when a new sample is sent to the network, the self-supervised learning model is used to embed the features of the sample, and then the anomaly score is calculated in the whole image and the small patch to achieve anomaly detection and localization. Zavrtanik et al. [19] proposed an anomaly detection method based on image reconstruction of generative adversarial network, which first using the CutPaste strategy to construct a batch of negative samples from positive samples, and then constructs negative samples by generating adversarial network and compares them with positive samples to get the defect localization.

The methods of anomaly detection and image reconstruction can determine whether the image contains defects, but the accuracy of defect localization is unsatisfactory. Therefore, in order to achieve high-precision defect localization, the method based on detection and segmentation has gradually become the mainstream. HMFCA-Net [2] proposes an attention mechanism using multi-frequency information and local cross-channel interaction to represent weighted defect features, and introduces a ResNeSt network based on deformable convolution to deal with various defect shapes. Zhu [7] et al. introduced the transformer encoder block [9] and convolutional block attention module (CBAM [24]) into the PANet [6], which solved the problem of small object detection in dense and overlapping situations. Lin et al. [20] propose a framework for detecting image defects of LED chips based on AlexNet architecture, by removing Alexnet's fully connected layer and adopting class-activation maps (CAMs), which allows them to learn using only per-image labels and using CAMs for the localization of the defects. However, the image resolution available for this method is low. Tabernik [21] proposed a two-stage defect detection model by combining the segmentation model with the decision model, the segmentation network performs pixel-level localization of defect detection, treating each pixel as an individual training sample, while the decision network uses the segmented features as input to judge the class information of the segmentation block, but this two-stage approach does not allow end-to-end training[22]. proposed a two-stage end-to-end defect detection algorithm, which is achieved by balancing the contribution of segmentation and classification loss, and adjusting the gradient flow from classification to segmentation network. Although these methods have got impressive results in defect detection, their performance in small defect detection needs to be improved.

This paper focuses on improving the performance of defect detection algorithms on small defect detection, and proposed a Multi-scale Attention and Dilation Network for small defect detection, which will be introduced in the next section.

3. Multi-scale attention and dilation network

This section will introduce our proposed small object detection method. In Section 3.1, we will introduce the proposed multi-scale attention and dilation network. In Section 3.2, we will explain how to build high-resolution feature maps. In sections 3.3 and 3.4, we will introduce the attention module and the context aggregation module, respectively.

3.1. Overview

Motivated by the idea that the feature layer with higher resolution has a greater contribution to the detection of small objects, we hope to build a feature layer with higher resolution. Higher resolution will reduce the perceptual field of the feature layers and lose the necessary global information for object detection, the global information often contains contextual information about the object. We propose a new feature aggregation network to improve the performance of small defects detection by introducing attention mechanism and context aggregation module. We call the proposed feature aggregation network as Multi-scale Attention and Dilation Network (MADNet). The overall architecture of our multi-scale dilated feature layer is shown in Fig. 1, C2 to C5 are features derived from the backbone.

3.2. High-resolution feature maps

The features obtained by downsampling factors 4,8,16,32 of the backbone are called C2, C3, C4, C5, and in YoloV5 [25], PANet [6] utilizes C3, C4, C5 to build a multi-scale feature layer which is called P3, P4, P5. We find that the contribution of P3 to the detection of small objects is much greater than P5, that is, the higher the resolution of the feature layer, the better the recognition ability of small objects. The reason is obvious: the large perceptual field of the P5 feature layer is difficult to perceive the information of small objects. Therefore, we build a multi-scale feature layer with higher resolution through PANet on the resolution of the three feature layers C2, C3 and C4 of the backbone, and use the prediction head to make regression prediction on the constructed high-resolution feature layers P2, P3, P4. The C5 feature layer has little contribution to small object detection, so no prediction head is added to the C5 feature layer in our feature aggregation network. However, the high-level feature information contained in the C5 feature layer is helpful for object detection, so we use the top-down and bottom-up path of PANet [6] to aggregate the information of the C5 feature layer into P2, P3, and P4 to enrich their feature expression.

3.3. Attention module

For deep features such as C5, due to the large perceptual field, the local information of small objects is often not well perceived.

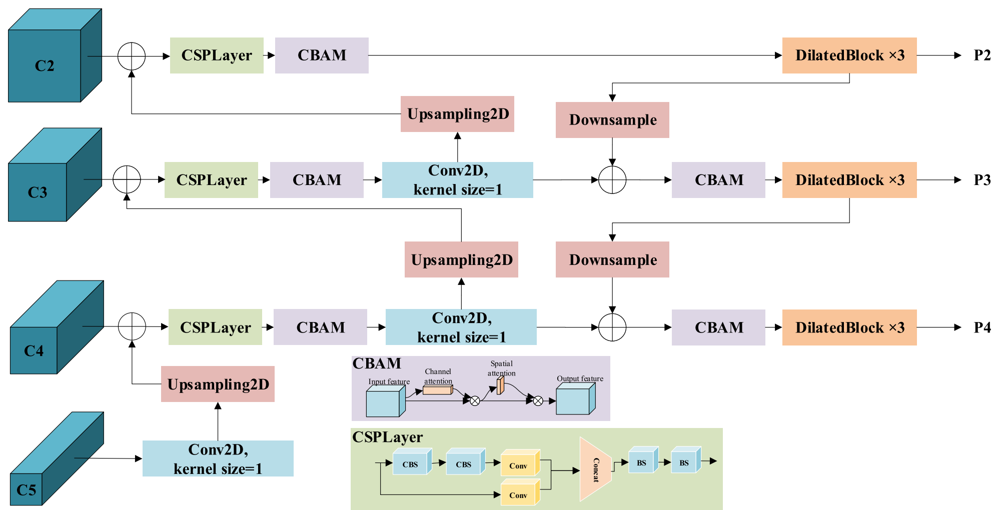

Fig. 1. The structure of multi-scale attention and dilation network.

In the feature aggregation network of yolov5 [25], the deep features are directly aggregated to shallow features, which may not be helpful for the detection of small objects. So after feature fusion at different scales, a convolutional block attention module (CBAM [24]) is used to strengthen the local features of small objects in this paper. CBAM [24] contains 2 independent submodules, Channel Attention Module (CAM) and Spartial Attention Module (SAM), which perform channel and spatial attention respectively. Channel Attention Module (CAM) calculates the importance of different feature maps and assigns different weights to feature maps, and Spartial Attention Module (SAM) calculates the weights of different pixels on the feature map and assigns different weights to pixels. A larger pixel weight means that the more important the pixel is in the feature map, the more likely it is to be a feature useful for the detected object.

3.4. Context aggregation module

Increasing the resolution of the feature layer means that its perception field will become smaller, which will make the network lose the necessary global information for object detection, making the model detection effect worse. In order to alleviate the problem of limiting the perception field of high-resolution feature layers, we propose a context aggregation module. We use the stacking of dilated residual blocks with different dilation rates to improve the perception field under the premise of ensuring the resolution of the feature layer. The dilated residual block first adjusts the number of channels of the feature map by 1×1 convolution, then increase the perception field of the feature map through a dilated convolution, and finally restores the channel number to the same as the input by 1×1 convolution and add the input at the pixel level, as shown in Fig. 2. The number of channels and resolution of the output is the same as the input, but the perception field is increased by dilated convolution, the dilated convolution can be set to different dilation rate. At the same time, because the kernel of dilated convolution is not continuous, that is, not all pixels will be calculated, which will ignore some point information, which is called grid effect [26]. Referring to the HDC principle [26], a different rate is used for each dilated residual block in this paper, and the dilation rate is transformed into a serrate form, that is, dilation between different layers is constantly changing, with the goal of the final receiving field covering the entire area. Specifically, we get three feature layers of P2, P3, and P4 through PANet [6], with resolutions of 160, 80, and 40, respectively, then we stack three dilated residual blocks with dilation ratios [1,2,3] behind each feature layer to expand the perceptual field and obtain the final multi-scale feature layers. It is worth nothing that we also use CSPLayer in our feature aggregation network to enhance the aggregation capabilities of features, but unlike the CPSLayer_N in backbone, CSPLayer only stacks two CBS blocks without residual connections.

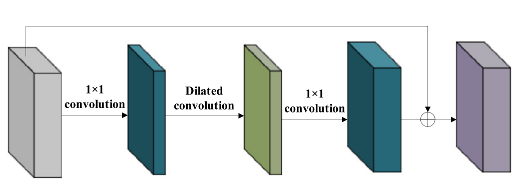

Fig. 2. The structure of dilated residual block.

More intuitively, the dilated residual block here is actually aggregating the context information of the small object. Specifically, if we get the features with a resolution of 160×160 through backbone, each pixel is a local fine feature of a small object obtained on a small perceptual field, and there is an dilated convolution with an dilation rate of 1 and a convolution kernel of 3×3, the global context information adjacent to each pixel (if the convolution kernel is 3×3, the global context information of 8 pixels will be aggregated) can be aggregated on a new feature map. The feature map after aggregating the context not only has the small object information extracted locally, but also contains the context information globally, which has a stronger representation ability for small objects. To put it simply, our feature aggregation network not only obtains more local and finer features of small objects on a smaller perceptual field, but also aggregates global contextual information through dilated blocks by expand the feature-layer perceptual field. Finally, the overall architecture of our multi-scale dilated feature layer is shown in Fig. 1, C2 to C5 are features derived from the backbone. Conv2D halves the number of channels, CSPLayer is consistent with YoloV5, and the three dilated blocks are stacked with dilation rate of [1,2,3].

4. Experiments

We choose YoloV5 [25] as our baseline. Like SSD [27], YoloV5 [25] is a classic one-stage detector. The advantage of one-step detectors over two-step detectors is the fast speed of network inspection. All experiments in this paper use CSPDarknet53 [28] as the backbone, we evaluate our model on the DAGM 2007 [11] and In-

Table 1
Comparison results on DAGM 2007 for each category.

Model	mAP(0.75)	Class1	Class2	Class3	Class4	Class5	Class6	Class7	Class8	Class9	Class10
YoloV5_s	70.30	48.90	59.66	69.79	87.99	87.00	72.92	49.14	68.56	91.20	65.13
YoloV5_m	74.17	72.90	65.77	75.14	94.18	95.00	75.00	36.85	61.46	91.50	73.88
YoloV5_l(baseline)	76.56	65.41	73.81	62.59	89.47	84.00	85.78	49.96	86.38	91.81	76.35
MADNet(ours)	77.65	73.55	74.46	77.85	81.72	97.49	60.94	48.00	87.11	92.03	83.30

dustrial Surface Defect Dataset. We compare our model with baseline Yolov5 [25] and analyze the advantages and disadvantages of our model.

4.1. Implementation details

DAGM 2007 [11] has a image resolution of 512 × 512, and there are ten classes, where the first six classes have a sample size of 150 images, last four classes have a sample size of 300 images, and the dataset has a total of 2100 images. In order to evaluate the model effect more effectively, we divide the dataset into training and test sets according to 9:1, and report the final results on the test set. We train our model with synchronized SGD [29] with a total of 16 images per mini-batch. Initial learning rate is 0.01 and 512 × 512 input for DAGM 2007 and 960 × 960 for Industrial Surface Defect Dataset. We use CSPDarknet53 [28] pre-trained model for backbone to make the model converge faster during training. The parameters of the feature aggression network are randomly sampled from a normal distribution with a mean of 0 variance of 0.02. We first freeze the backbone network to train 50 epochs, and then unfreeze the backbone network to train 150 epochs. Following the HDC principle [26], the dilation rate of the dilated blocks is set to [1,2,3] in all experiments except the ablation experiment in this paper. For the rest of hyperparameters, we refer to YoloV5 [25].

4.2. Comparison results on DAGM 2007

We trained our model with input images with 512 × 512 resolution, and we train the model on the divided training set and verify the results on the test set. The Yolov5 comes in three different sizes, namely Yolov5_s, Yolov5_m, and Yolov5_l, which control the number of parameters and calculations of the model by the number of residual structures of the model. Yolov5_l has the largest amount of network parameters and calculations, but the highest accuracy. We use the backbone of yolov5_l, and for a fair comparison, we choose yolov5_l as our baseline. Comparison results are shown in Table 2. In Table 2, AP0.75 is the mAP when the IOU threshold is 0.75, AP0.5:0.95 is the average value of each mAP when the IOU threshold is taken from 0.5 to 0.95 in steps of 0.05. S, M, and L represent mAP on small, medium, and large objects, respectively. We refer to the MS COCO [10] to define the object size: pixel area less than 32 square is a small object, less than 96 square is a medium object, and more than 96 square is a large object. We calculate the mean AP values of 0.5 to 0.95 at 0.05 steps on small, medium, and large objects. More detailed results on each class of DAGM 2007 are shown in Table 1.

Table 2
Comparison results on DAGM 2007.

Models	Parms(M)	FPS	AP0.75	AP0.5:0.95		
				S	M	L
YoloV5_s	7.3	83	70.30	65.1	58.8	56.5
YoloV5_m	21.376	50.46	86.5	66.4	63.9	60.7
YoloV5_l(baseline)	47.05	46.9	76.56	67.0	65.2	65.9
MADNet(ours)	48.38	45.3	77.65	72.8	66.7	62.8

Since the dilated blocks increased the perception field and enables the aggregation of contextual information of small objects, our method improves the mAP of YOLOV5_l on small objects and medium objects by 5.8% and 1.5%, and AP0.75 increased from 76.56% to 77.65%. At the same time, dilated blocks only brings a small increase in computational overhead. From the results of Table 2, it can be seen that although our model has obvious advantages in the detection of small and medium-sized defects, its detection performance on large defects is not as good as YoloV5. The reason for this result is obvious, we increase the resolution of the feature layer, the small receptive field of the high-resolution feature map is beneficial for small defects detection, but not for large defects. That is, our method is more suitable for extracting more local, fine-grained features and expanding the perceptual field by aggregating contexts, but the large objects, on the other hand, require a larger field of perception to perceive more useful information. So the rensons above lead to our model in the small defects detection performance is improved, but the large defect detection performance is reduced.

We visualize some attention on the DAGM 2007 in Fig. 3 to observe the area of interest of our method when detecting defects after adding the CBAM [24], and compare it with the attention of YoloV5. This paper uses Grad-CAM [30] to get the attention map. As we can see from the Fig. 3, our method has smaller regions of interest due to the higher resolution of the feature layer and the smaller perceptual field, but this can be compensated by aggregating the object context by dilation residual blocks, which is beneficial for the detection of small defects. Meanwhile our method can focus on the defects to be detected compared to YoloV5.

4.3. Comparison results on industrial surface defect dataset

We collect a 500-image Industrial Surface Defect Dataset with two types of defects: scratch and spot. The resolution of the images we collect is 1024*1024, and during training, we resize the images to 960 × 960. We use 400 of them as the training set and the remaining 100 as the test set. There are a total of 4275 scratch instances and 2630 sopt instances. We train our model for defect dataset with synchronized SGD with a total of 8 images per minibatch, and the initial learning rate of 0.01.

The comparison results with more models are shown in Table 3. It can be seen in Table 3 that our method achieves the best results on Industrial Surface Defect Dataset compared to mainstream detection algorithms. The mAP improved by 0.9%, APs increased by 2.6%, APm increased by 3.9%, and APl decreased 2% compared to Yolov5. This is consistent with the results in DAGM 2007. The reasons have already been analyzed in Section 4.2. We will take how to ensure the performance of large defects detection on the basis of improving the detection performance of small defects as our future work. Some visualization results from our method on the Industrial Surface Defect Dataset are shown in Fig. 4.

When the dataset sample is limited, there is a great risk of over-fitting the model, resulting in a large performance difference between the simulation benchmark and real application. To making our experimental results more convincing on the Industrial Surface Defect Dataset, we do a 5-fold cross-validation, specifically, we divide the Industrial Surface Defect Dataset into 5 subsets, each sub-

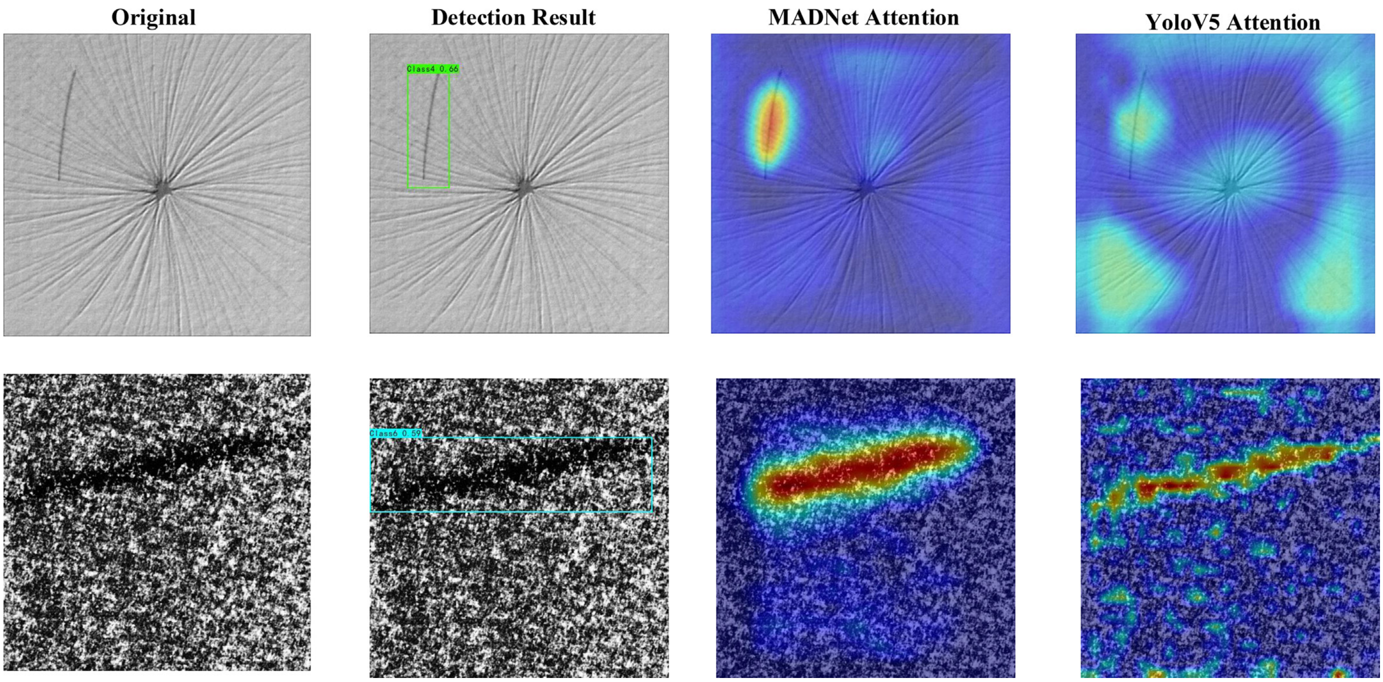

Fig. 3. Some visualization of attention between our method and our base method YoloV5 on the DAGM 2007.

Table 3
Comparison results on Industrial Surface Defect Dataset.

Methods	mAP0.5	mAP0.75	APs	APm	APl	mAR0.5:0.95	ARs	ARm	ARl
Faster RCNN	83.9	36.2	19.2	42.0	51.9	51.5	20.0	52.4	58.8
Cascade RCNN	85.8	39.0	14.8	44.6	55.6	54.4	17.3	55.0	62.3
ATSS	81.9	36.4	14.8	41.5	55.8	53.6	14.6	55.0	64.0
YOLOv3	70.5	17.3	5.1	28.7	36.1	38.2	5.4	39.8	43.9
YOLOX	78.3	29.6	11.8	38.6	43.0	49.3	14.4	50.4	55.7
RetinaNet	78.0	29.4	9.6	37.6	44.9	50.0	19.2	51.8	56.1
EfficientNet	84.3	34.2	13.9	41.8	51.9	53.4	25.2	53.4	63.0
Defomable-DETR	79.7	27.5	9.7	37.6	46.2	50.1	11.0	51.2	58.6
Swin Faster RCNN	86.5	40.3	6.0	45.9	54.9	51.6	6.0	53.8	61.9
Faster RCNN Res2Net-101	85.5	38.4	13.3	44.7	54.8	52.5	14.2	53.3	61.4
YoloV5_s	80.5	22.1	7.1	32.8	50.9	46.0	17.8	45.3	62.2
YoloV5_m	84.71	27.3	9.2	36.9	53.1	48.8	14.1	48.6	65.0
YoloV5_l(baseline)	88.3	32.8	19.9	41.8	57.1	50.5	27.7	50.6	65.0
MADNet(ours)	89.2	39.7	22.5	45.7	55.1	50.6	28.4	50.9	63.0

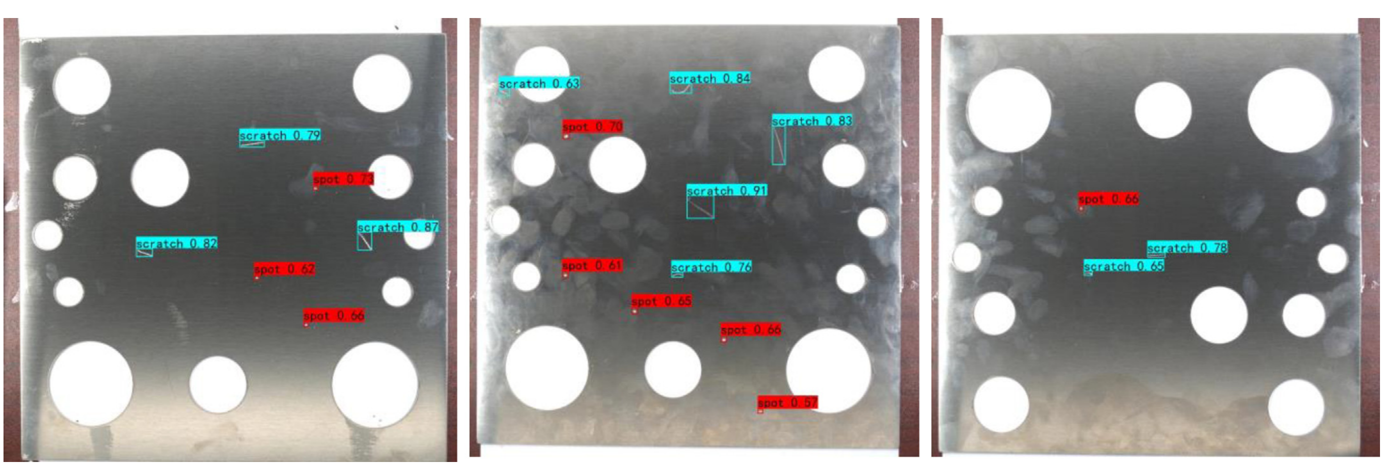

Fig. 4. Some visualization results from our method on the defect dataset.

Table 4
The results of cross-validation.

Data	mAP(0.5)	APs	APm	APl
Subset1	85.9	15.6	39.2	48.7
Subset2	86.5	18.7	40.9	51.7
Subset3	88.3	21.8	43.1	54.9
Subset4	87.3	19.8	42.9	53.2
Subset5	86.4	20.1	41.8	50.3
Entire dataset	86.9	19.2	41.6	51.8

Table 5
Comparison results on different FPN variants.

Model	mAP(0.5)	APs	APm	APl
Backbone+FPN	86.4	17.2	40.2	54.7
Backbone+PAFPN	88.3	19.9	41.8	57.1
Backbone+HRFPN	88.7	20.4	42.1	56.9
MADNet(ours)	89.2	22.5	45.7	55.1

set contains 400 training samples and 100 test samples, and ensure that the test samples of each subset have no intersection, then we train and evaluate the model on each subset separately, and finally report the average of the evaluation results on the five subsets as the final model evaluation results. The results of cross-validation are shown in the Table 4.

4.4. Comparison results on different FPNs

There are already many effective FPN versions, which have got impressed results in object detection, however, most of them are through consideration to build complex and effective feature aggregation paths, or adopt different feature fusion methods, and few studies consider improving the components themselves used to build FPNs. This paper focuses on improving the components of the FPNs, constructing the attention components based on the CBAM [24] module which have channel attention and spatial attention, and constructing the context aggregation components based on the dilated convolutional block. We referring to the feature fusion path of the PANet [6] to construct our FPN, and comparing the performance of our FPN with different FPN variants under the same training strategies. The comparison results obtained are shown in Table 5. From the results in the table, we can see that by introducing attention and context aggregation, our FPN detection performance on small defects is significantly improved.

4.5. Ablation studies

Our method contains two crucial components: the CBAM attention module and the dilation residual blocks. To verify if they really work in our method, we perform ablation experiments, all of which are trained and tested on Industrial Surface Defect Dataset.

Dilation or not To verify the necessity of dilated convolution blocks in our multi-scale dilated feature layer, we perform ablation experiments. Under the premise of ensuring that the network structure remains unchanged, we only remove the dilated convolutional blocks. The comparison results obtained are shown in Table 6. From the results in the Table 6, it can be seen that without dilated convolution blocks, the performance of the detector on different sacles objects decreases significantly, and the mAP is decreased by 3.1%. This shows that after the high-resolution feature layer extracts the fine local features of small defects, it is necessary to aggregate the context information of small objects by expanding the receptive field for small defects detection. So the dilated convolutional blocks are crucial component of our model.

Dilation rate We stack three dilated convolutional blocks to expand the feature layer perception field for context aggregation of small objects. The dilation rate of dilated convolution affects the ability of the model feature aggregation context and directly determines the performance of the model, so it is particularly critical to choose the appropriate dilation rate. If the dilation rate is increased exponentially, some information will not be used, the loss of information is detrimental to small object detection, and setting a serrated dilation rate can avoid appeal. To verify the effectiveness of the serrated dilation rate, we change the dilation rate of the dilated convolution and observe the experimental results. The comparison results obtained are shown in Table 7. It can be seen that the serrated dilated rate has the best results.

Attention or not In this paper, we utilize the convolutional block attention module (CBAM [24]) to add attention to the feature aggregation network.This is to allow the deep features to make better use of the information of small objects after being aggregated into shallow features, which enhances the feature fusion capability. To verify that the attention module works in our feature aggregation network, we remove the CABM [24] from the network and leave the other components unchanged. The comparison results obtained are shown in Table 8.

Finally, we give the result of removing both the CBAM attention module and the dilation residual blocks, as shown in Table 9. From the results of ablation experiments, we can see that both the CBAM attention module and the dilation residual blocks are crucial components for our method: one to make the network features better focus on the object features, and the other to aggregate the object context, removing these components will lead to a significant decrease in the performance of the network.

Table 6
Ablation experiment results on dilated convolution.

Models	AP0.5	AP0.5:0.95		
		S	M	L
MADNet	89.2	22.5	45.7	55.1
MADNet without dilated convolution	86.1	20.1	42.6	53.7
YoloV5	88.3	19.9	41.8	57.1

Table 7
Ablation experiment results on dilation rate.

Models	AP0.5	AP0.5:0.95		
		S	M	L
1,1,1	85.9	20.1	43.7	50.7
2,4,6	86.2	21.3	43.7	51.3
1,2,3	89.2	22.5	45.7	55.1

Table 8
Ablation experiment results on CBAM.

Models	AP0.5	AP0.5:0.95		
		S	M	L
MADNet	89.2	22.5	45.7	55.1
MADNet without CBAM	87.5	21.4	43.1	52.9
YoloV5	88.3	19.9	41.8	57.1

Table 9
Ablation experiment results on dilated convolution and CBAM.

Models	AP0.5	AP0.5:0.95		
		S	M	L
MADNet	89.2	22.5	45.7	55.1
MADNet without dilated convolution and CBAM	79.7	12.1	33.6	50.2
YoloV5	88.3	19.9	41.8	57.1

5. Conclusion

In this paper, a networ for small defect detection is proposed. We get better detection performance on small defect by increasing the resolution of the feature layers, and spatial and channel attention is introduced to improve the network's perception of small defects, and we using dilation residual blocks to solve the problem of limited perceptual field brought about by increasing resolution, which aggregates the global context information of small defects. We call our method MADNet. Our MADNet not only obtains more local and finer features of small defects on a smaller perceptual field, but also aggregates global contextual information through dilation residual blocks by expanding the feature-layer perceptual field. Our experiments show that our method is superior to advanced detectors such as YOLOV5 in small defects detection. But our method has a slight degradation in detection performance on large defects. We will consider how to ensure the performance of large defects on the basis of improving the performance of small defects as future work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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